DIY information provision with mobilityHubTools: CERC presentation notes

James Reynolds

2026-10-17

- *Situation:* Shared mobility hubs[1] are a (relatively) new direction in transport. They co-locate shared, micro-mobility transport modes (sometimes with electrified and/or public transport services or other facilities[2]). This might provide options for (more occasional) trips that are difficult on foot, bicycle or using transit, and so contribute to more sustainable travel patterns.[3]
- *Problem:* Not much is yet known about providing information at mobility hubs. While there is some guidance from CoMoUK and others, there do not appear to be tools available to support practitioners implementing hubs. Rather, mobility hub information provision appears to be largely done on a site-by-site, system-by-system or otherwise bespoke basis.
- *Question:* Is it possible to improve understandings of current (and best) practices, and to develop tools to support implementers?
- *Answer:* Undertake design and case-based research guided by the iterative Double Diamond process. First, examine selected mobility hubs and *discover* current practices. Then *define* what mobility hub information provision might be trying to solve, *develop* design solutions and *deliver* a package of software.

[1] "...a place where different transport modes are integrated seamlessly, promoting efficient and sustainable urban mobility. Emphasis is given to shared mobility options such as bikes, scooters and cars. Connection with public transport (transit) is desirable but not necessary. The integration between mobility suppliers is important to provide a seamless, flexible connection at these transfer points. Ensuring an enjoyable experience for travelers is a crucial part of the hub concept as well, and therefore the hub should be considered not just as a transfer node but as a place where a variety of services come together, such as logistics services or restaurants." (Roukouni et al. 2023)

[2] Anastasia Roukouni et al., "An Analysis of the Emerging 'Shared Mobility Hub' Concept in European Cities: Definition and a Proposed Typology," Article, *Sustainability (Switzerland)* 15, no. 6 (2023), doi:10.3390/su15065222.

[3] Karst Geurs et al., "The Smarthubs Integration Ladder: A Conceptual Model for the Categorisation of Shared Mobility Hubs," *Transport Reviews* 44, no. 1 (2024): 112–39, doi:10.1080/01441647.2023.2239499.

Background

Draft intro video
https://www.youtube.com/watch?v=hu9jsgi8CVU

Draft mobility hub website
https://james-reynolds.github.io/ROBUST_Electric_shaRed_mOBility_hUbS_Trial_website/

Draft mobility hub website, Github source code
https://github.com/James-Reynolds/ROBUST_Electric_shaRed_mOBility_hUbS_Trial_website

Figure 1: Draft mobility hub video and website links

This research is part of the Electric shaRed mOBility hUbs Trial (ROBUST) project[4]. The project includes co-design, behaviour nudg-

[4] See `https://www.tcd.ie/transport-research/our-research/research-projects/ROBUST-electric-shared-mobility-hubs-trial/`

ing and hub incubation activities.

The literature suggests: branding, signage, guidance, 'totem' displays, and online information[5].[6] It also notes a need for: "integration", "seamless, flexible connection" and ". . . an enjoyable experience for travelers" (Roukouni et al. 2023).

[5] CoMoUK, *Mobility Hub Toolkits*, 2021, https://tinyurl.com/ux2svctn.

[6] CoMoUK, *Communications: How to Increase Awareness and Support for Mobility Hubs*, 2021, https://tinyurl.com/5cxuxeru.

Figure 2: CoMoUK increasing awareness case study. Source: CoMoUK (2021)(6)

Methodology

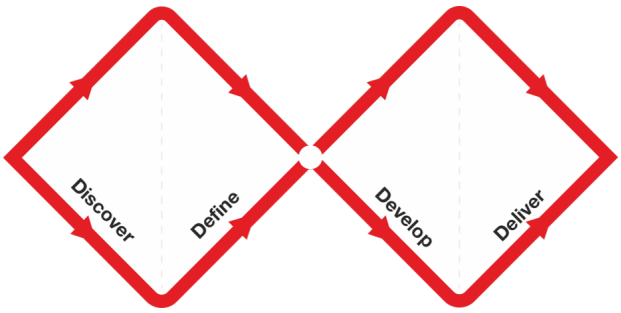

Figure 3: Framework for Innovation. Source : UK Design Council (undated)

- industrial design research approach, apllicable to transport,[7]
- alternate between divergent and convergent approaches,
- overall process is iterative.

[7] Selby Coxon, Robbie Napper, and Mark Richardson, *Urban Mobility Design* (Elsevier, 2018), doi:10.1016/C2017-0-01129-1.

Discovery and definition using case-based research approach:

- effective for early stage research, and for 'how' or 'why' questions[8].
- select cases for theoretical reasons (e.g. leading case, representative case, critical case, or for polar opposites (cross-case comparison))(Robert Yin[9] and others)[10].
- duality criterion: grounded within the context of the small number of cases (for depth of research), yet at the same time seeking generalisable results that apply to more than just the cases studied (Martyn Denscombe[11] and others).

Development and delivery: iterative coding to replicate various examples; develop synthesis version (industry practice + respond to defined problem); and deliver as a software package.[12]

Results

- *Discovery:* wide range of examples[13].

- *Definition* of the problem to be addressed by the design output:

  1. What is this?
  2. Where am I?
  3. What is here?
  4. What is nearby?
  5. How can I travel from here?
  6. (*new*)[14] How do I use each bit of this?.

- *Development* of: Bristol-style totem, Birmingham-style totem and 'synthesis' totem. Examples in Figure 5, but also up on the walls around us.

- *Delivery* of: R package on github[15]; code developed to produce paper outputs[16]; and these notes and the associated outputs used in this presentation[17].

Discussion and conclusions

- Tactical urbanism: why not make a "mobility hub" of your own?
- More to do to improve the code, develop further outputs etc. (Open source, so you can help with this).
- Open Street Maps data quite powerful, as are the GIS tools available for R.

[8] How is information provided at and about mobility hubs? How might information at and about mobility hubs be provided ideally? How can such best practices can be made generally applicable?

[9] *Case Study Research and Applications: Design and Methods*, Sixth (Thousand Oaks, CA, USA: SAGE Publications, Inc., 2018).

[10] Cases: (*Redbridge*) South Woodford and High Street, Wanstead; (*Camden*) Goldington Crescent, and Doric Way; (*Bristol*) Abbey Wood, UWE Frenchay, and Gainsborough Square; and (*Birmingham*) Huntingtree Park and Andrew Road

[11] *The Good Research Guide*, 2nd ed. (Maidenhead, Berkshire, UK: McGraw-Hill International UK Ltd, 2007).

[12] Hadley Wickham and Jennifer Bryan, *R Packages* (" O'Reilly Media, Inc.", 2023), `https://r-pkgs.org/`.

[13] `https://maps.app.goo.gl/1dd8sEJCiwCgPkc98`, `https://maps.app.goo.gl/j7b7MLr8HzcjqoS66`, `https://maps.app.goo.gl/BW8cEVXioYSoNbiz9`, `https://maps.app.goo.gl/1JqvPLWkwkbnyGB27`

Figure 4: Selected hub images. Source: authors.

[14] This is coming out of subsequent co-design workshops.

[15] `https://github.com/James-Reynolds/mobilityHubTools`

[16] `https://github.com/James-Reynolds/WP5_mobilityHubTools_papers/blob/main/ReynoldsEtAl2026_CERC.Rmd`

[17] `https://github.com/James-Reynolds/WP5_mobilityHubTools_papers/blob/main/ReynoldsEtAl2026_CERC_preso.Rmd`

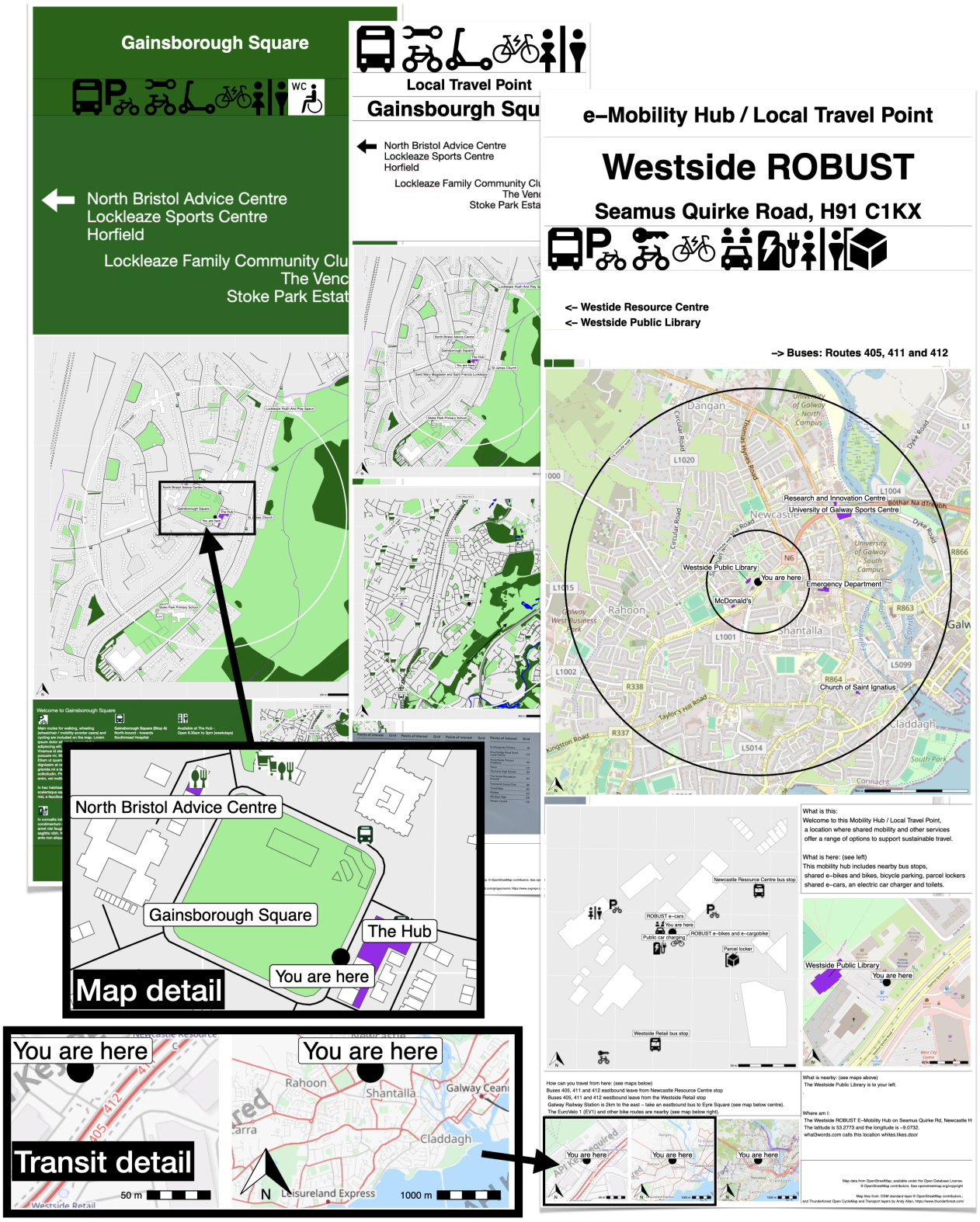

Figure 5: Software output examples: Bristol, Birmingham and synthesis-style totem displays (left, centre, right). Source: authors.